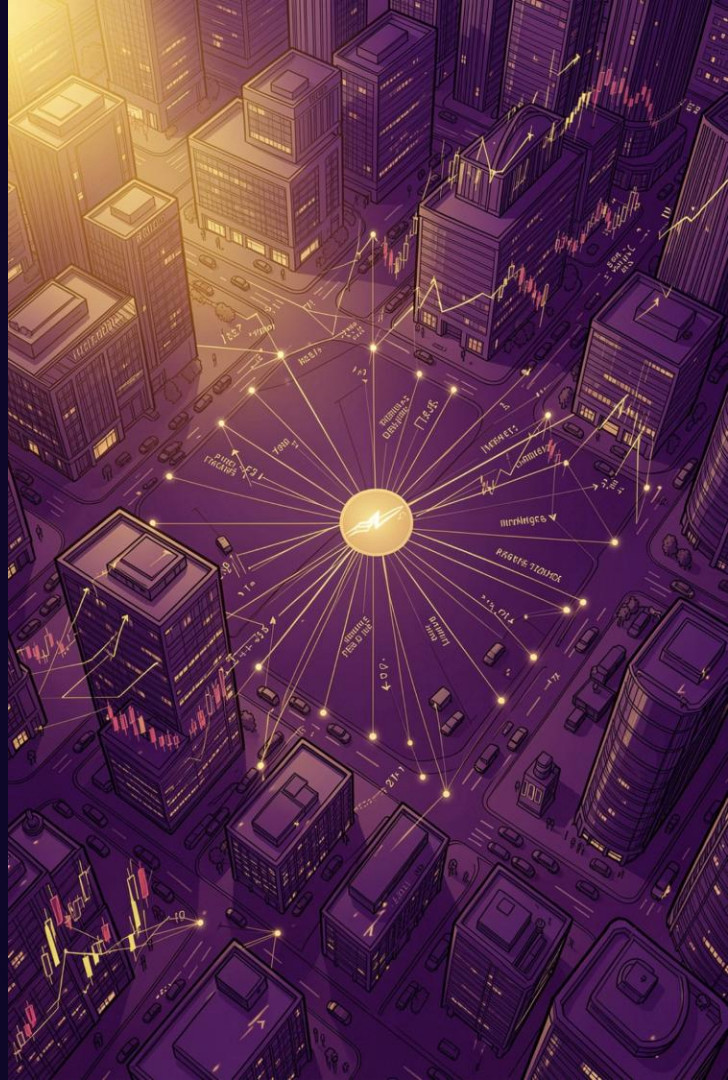
AML Detection Pipeline — End-to-End ML

PaySim dataset → Preprocessing → PCA → K-Means behavioral segments → XGBoost classifier → Continuous (incremental) learning



Toolbox

Data & Math

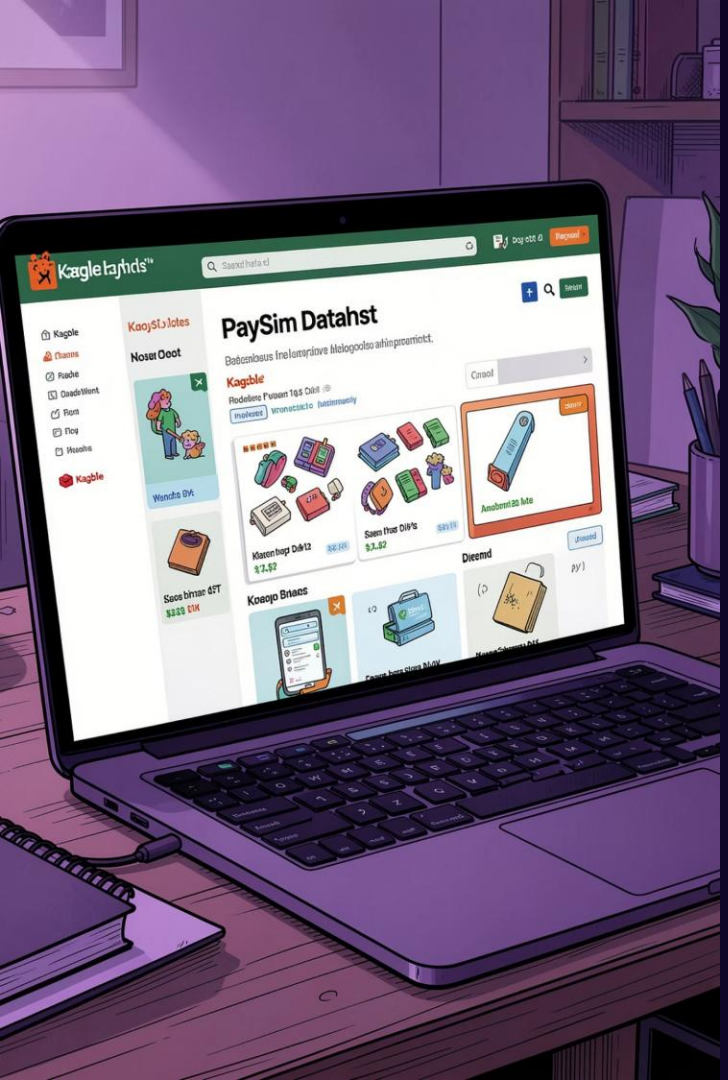
pandas, numpy — table and numeric operations

Visualization

matplotlib / seaborn — exploration & diagnostics

Modeling

sklearn (PCA, KMeans, scaling) + xgboost classifier



Data

PaySim (synthetic financial transactions). Save as paysim.csv in the working folder. Inspect shape, class distribution, and fraud rate before modeling.

Phase 1 — Preprocessing & Feature Engineering

Drop

Remove nameOrig, nameDest, isFlaggedFraud

Missing Values

Check and impute if present (PaySim expected clean)

Encode

Label-encode transaction type (TRANSFER, CASHOUT, etc.)

Scale

StandardScaler → mean=0, std=1

Phase 2 — PCA (Dimensionality Reduction)

Phase 3 — Behavioral Segmentation (K-Means)

Cluster transactions in PCA space to reveal behavior patterns. Use the Elbow Method (k=2..10) to pick k. The cluster label becomes behavioral_segment — a powerful engineered feature for XGBoost.

ⓘ Clusters often separate fraudulent behavior from normal patterns.

Phase 4 — XGBoost Fraud Classifier

Train/Test Split

80% train / 20% test, stratify to preserve fraud ratio

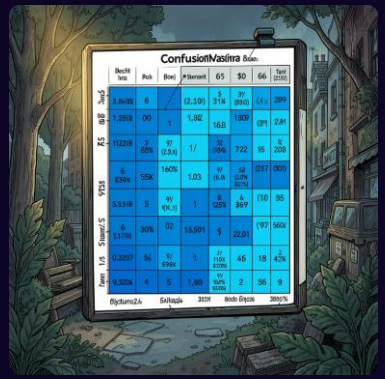
Handle Imbalance

scale_pos_weight = normal_count / fraud_count

Evaluate

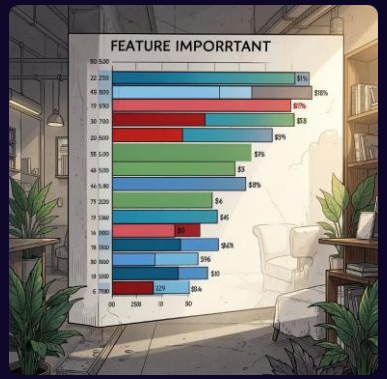
Use Precision, Recall, F1 — NOT accuracy

Model Diagnostics



Confusion Matrix

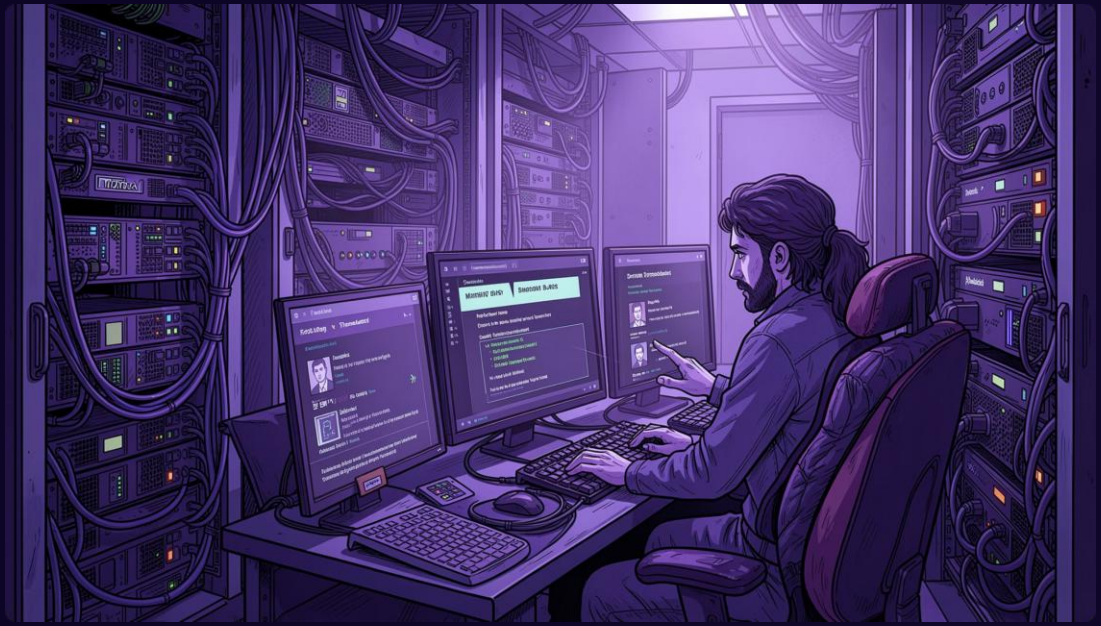
Shows true negatives, false positives, false negatives, true positives — key for understanding errors.



Feature Importance

Reveals which features (including behavioral_segment) the model relied on most.

Phase 5 — Continuous Learning (Incremental Update) Update)



Simulate new-month data as a new batch. Rule: do NOT mix with old data. Use XGBoost warm-start (xgb_model / booster) to continue training on new data only (e.g., +20 rounds).

- Adapts to evolving fraud tactics without storing all historical data.

Executive Summary & Q&A

Business Problem

Balance catching money laundering while minimizing false alerts that overwhelm analysts.

5-Phase Solution

Preprocessing → PCA → K-Means → XGBoost → Continuous learning (incremental updates).

Why Metrics Matter

Use Recall and F1-score instead of accuracy because fraud is <1% of transactions.